

Assume then (A1)

(I65)

Ideally, we would want $\hat{\underline{\theta}}^* = \underline{\theta}$ (A2)

But we are in general not able because of lack of data.

Data Set = $\{ u(k), y(k), k = 1 \dots N \}$

If N is "small" $\Rightarrow$ noise can be "dominant" and we could not be able to obtain a "good" $\hat{\underline{\theta}}$

In principle, we want

$\hat{\underline{\theta}}^*$ obtained minimizing some "cost function"
possibilities $\rightarrow$ min "in average" ~~the~~ the expected prediction error $\leftarrow$
 $\rightarrow$ min. the maximum possible error.

We want $\hat{\underline{\theta}}^*$ so that $[y(k) - \hat{y}(k|\hat{\underline{\theta}})]^2$ is minimum "in average"

$$\Rightarrow \hat{\underline{\theta}}^* = \arg \min_{\hat{\underline{\theta}}} E \{ [y(k) - \hat{y}(k|\hat{\underline{\theta}})]^2 \}$$

But, we are not able to compute the expectation.

In fact we have only a collection of IN/OUT data (Data Set)

we do not have any p.d.f. $\Rightarrow E\{\}$ $\rightarrow$ Empirical ~~cost~~ ^{mean value}

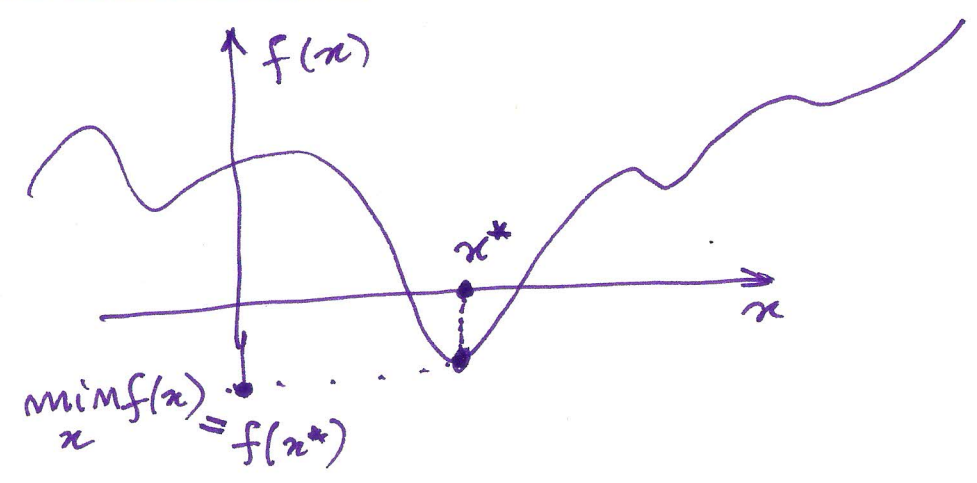
$$E\{[y(k) - \hat{y}(k|\hat{\theta})]^2\} \longrightarrow \frac{1}{N} \sum_{k=1}^N [y(k) - \hat{y}(k|\hat{\theta})]^2$$

$\uparrow$
Probabilistic.
expected value

$$\Rightarrow \hat{\theta}^* = \arg \min_{\hat{\theta}} \underbrace{\frac{1}{N} \sum_{k=1}^N [y(k) - \hat{y}(k|\hat{\theta})]^2}_{J_N(\hat{\theta})}$$

LEAST
SQUARES
MINIMIZATION

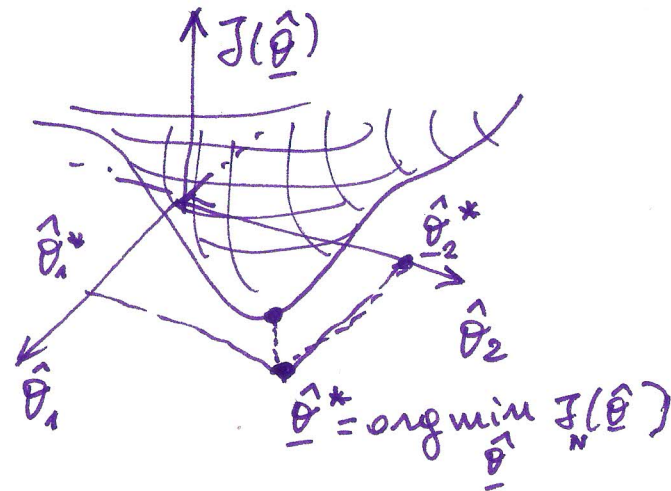
$x^* = \arg \min_x f(x) = ?$
(scalar case)



In our "multivariable" case

$$\hat{\underline{\theta}}^* = \arg \min_{\hat{\underline{\theta}}} J_N(\hat{\underline{\theta}})$$

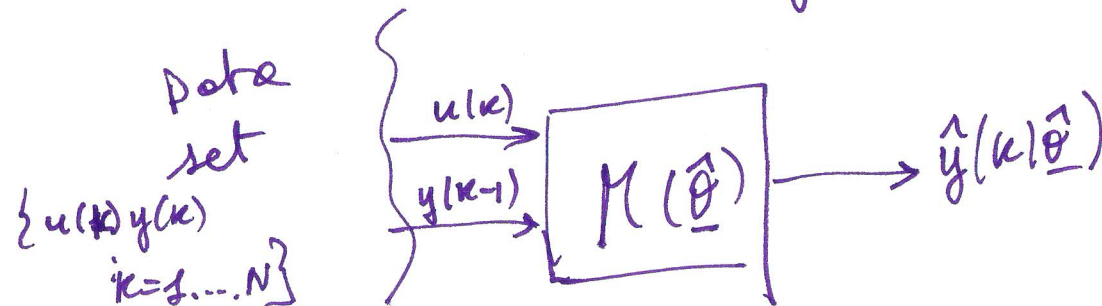
(assume $\dim \hat{\underline{\theta}} = 2$)



Note that

$$J_N(\hat{\underline{\theta}}) = \frac{1}{N} \sum_{k=1}^N \left[y(k) - \hat{y}(k|\hat{\underline{\theta}}) \right]^2$$

result of the application of the model
when parameters $\hat{\underline{\theta}}$ are used and
the sequence of inputs
 $u(1) \dots u(k)$ is assigned
and as measurement
 $y(1) \dots y(k-1)$ are used



General B.B. model

(I68)

$$\begin{aligned} y(k) &= G(z) u(k) + H(z) n(k) \\ &= \frac{B(z)}{A(z)} u(k) + \frac{D(z)}{C(z)} n(k) \end{aligned}$$

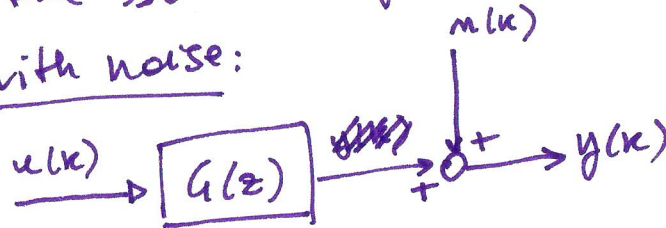
Particular cases:

1) Output Error Model

The Hp. is that the scheme of the system is:
ideal case:



with noise:



$n(k)$ ("the error") is a disturbance on the output.

We have

$$\begin{aligned} y(k) &= G(z) u(k) + n(k) \\ \boxed{y(k) &= \frac{B(z)}{A(z)} u(k) + n(k)} \quad (1) \end{aligned}$$

O.E. corresponds to the general case with $C(z)=D(z)=1$
($q_l=q=0$)

General predictive form for the System.

(168)

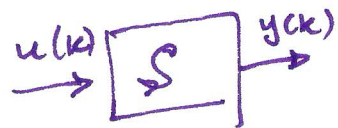
$$y(k) = \left[1 - \frac{C(z)}{D(z)} \right] y(k) + \frac{C(z) B(z)}{D(z) A(z)} u(k) + m(k)$$

(□) is already in pred. form. In fact

$$\begin{aligned} y(k) &= \underset{\substack{\uparrow \\ C(z)=D(z)=1}}{\left[1 - \frac{1}{1} \right]} y(k) + \frac{1 \cdot B(z)}{1 \cdot A(z)} u(k) + m(k) = \\ &= 0 + \frac{B(z)}{A(z)} u(k) + m(k) \end{aligned}$$

2) Equation Error Models

We could think of a difference eq.



Ideal case:

$$y(k) + a_1 y(k-1) + \dots + a_n y(k-n) = b_0 u(k) + \dots + b_m u(k-m)$$

$$y(k) + a_1 z^{-1} y(k) + \dots = b_0 u(k) + \dots + b_m z^{-m} u(k)$$

$$A(z) y(k) = B(z) u(k)$$

where $A(z) = 1 + a_1 z^{-1} + \dots + a_n z^{-n}$

$$B(z) = b_0 + b_1 z^{-1} + \dots + b_m z^{-m}$$

... (ideal case) ... $A(z) y(k) - B(z) u(k) = 0$

(I70)

In general, the latter is wrong.

$$\Rightarrow A(z) y(k) - B(z) u(k) = E(k)$$

$\uparrow$
error in the ^(difference) equation

$E(k)$ is not known $\Rightarrow$ Assumptions.

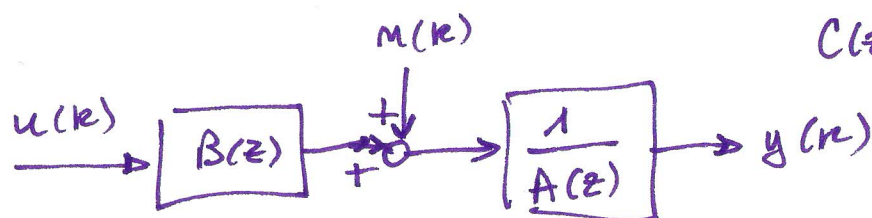
1) assume $E(k) = n(k) \sim WN(0, \lambda^2)$

$$\Rightarrow A(z) y(k) - B(z) u(k) = n(k)$$

ARX model

$$y(k) = \frac{B(z)}{A(z)} u(k) + \frac{1}{A(z)} n(k)$$

it corresponds to the gen. case with
 $D(z) = 1$
 $C(z) = A(z)$



Using the diff. eq. in k : $y(k) + a_1 y(k-1) + \dots + a_m y(k-m) = b_0 u(k) + \dots + b_m u(k-m) + n(k)$

A.R. X

$$y(k) = \underbrace{-a_1 y(k-1) - \dots - a_m y(k-m)}_{\text{Auto Regressive part}} + \underbrace{b_0 u(k) + \dots + b_m u(k-m)}_{\text{exogenous input}} + n(k)$$

This is a predictive form! $y(k) = \dots + n(k)$

from the general case:

$$y(k) = \underset{\substack{\uparrow \\ D(z)=1}}{\left[1 - \frac{A(z)}{1}\right]} y(k) + \frac{\cancel{A(z)} B(z)}{1 \cdot \cancel{A(z)}} u(k) + n(k) =$$

$$D(z) = 1$$

$$C(z) = A(z)$$

$$= \left[\textcircled{1} - A(z) \right] y(k) + B(z) u(k) + n(k)$$

$$\begin{aligned} & \parallel \\ & \left[\textcircled{1} - (1 + a_1 z^{-1} + \dots + a_m z^{-m}) \right] y(k) + \left[b_0 + b_1 z^{-1} + \dots + b_m z^{-m} \right] u(k) + n(k) \\ & \quad - a_1 \underbrace{z^{-1} y(k)}_{y(k-1)} - \dots - a_m \underbrace{z^{-m} y(k)}_{y(k-m)} + b_0 u(k) + b_1 z^{-1} u(k) + \dots + b_m z^{-m} u(k) + n(k) \\ & = -a_1 y(k-1) - \dots - a_m y(k-m) + b_0 u(k) + b_1 u(k-1) + \dots + b_m u(k-m) + n(k) \end{aligned}$$

- if (in the equation error model)

$$E(k) = D(z) n(k)$$

$n(k)$ is filtered by an FIR filter

↓
error is no more white but filtered $\Rightarrow$ coloured noise

$$A(z) y(k) = B(z) u(k) + \overbrace{D(z) n(k)}^{E(k)}$$

Coloured
Eq. Error
Model

$$y(k) = \frac{B(z)}{A(z)} u(k) + \frac{D(z)}{A(z)} n(k)$$

gen. case when
 $E(z) = A(z)$

or

$$y(k) = \underbrace{-a_1 y(k-1) - \dots - a_m y(k-m)}_{\text{Auto Regressive}} + \underbrace{b_0 u(k) + \dots + b_m u(k-m)}_{\text{exogenous input}} + \underbrace{n(k) + d_1 n(k-1) + \dots + d_q n(k-q)}_{\text{Moving Average of values of } n}$$

A.R.M.A.X

Predictive form.

$$y(k) = \left[1 - \frac{A(z)}{D(z)} \right] y(k) + \frac{A(z) B(z)}{D(z) A(z)} u(k) + n(k)$$

AR models.

$$A(z) y(k) = n(k) \quad (\text{no input})$$

ARX model and its identification.

(I-73)

$$y(k) = [1 - A(z)] y(k) + B(z) u(k) + n(k)$$

or, equivalently

$$y(k) = -a_1 y(k-1) - \dots - a_m y(k-m) + b_0 u(k) + \dots + b_m u(k-m) + n(k)$$

or,
$$y(k) = \underbrace{\varphi(k)^T}_{\varphi(k) = \underline{\theta}} \underline{\theta} + n(k)$$
 where
$$\underline{\theta} \triangleq \begin{bmatrix} a_1 \\ \vdots \\ a_m \\ b_0 \\ b_1 \\ \vdots \\ b_m \end{bmatrix} \quad \varphi(k) \triangleq \begin{bmatrix} -y(k-1) \\ \vdots \\ -y(k-m) \\ u(k) \\ u(k-1) \\ \vdots \\ u(k-m) \end{bmatrix}$$

$$\underline{a} \cdot \underline{b} = \underline{a}^T \underline{b} = [a_1 \dots a_p] \begin{bmatrix} b_1 \\ \vdots \\ b_p \end{bmatrix} = a_1 b_1 + \dots + a_p b_p$$
$$\underline{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_p \end{bmatrix}$$

Regressor Vector

ARX is linear in the Parameters

($y(k)$ is obtained by a lin. combin of the parameters weighted by the elem. in the regressor vector)

If we do not have $\underline{\theta}$, the model in ARX form will be

$$\begin{aligned}\hat{y}(k|\hat{\underline{\theta}}) &= [1 - A(z|\hat{\underline{\theta}})] y(k) + B(z|\hat{\underline{\theta}}) u(k) \\ &= [\hat{a}_1 z^{-1} - \dots - \hat{a}_n z^{-n}] y(k) + [\hat{b}_0 + \dots + \hat{b}_m z^{-m}] u(k)\end{aligned}$$

or

$$\boxed{\hat{y}(k|\hat{\underline{\theta}}) = \underline{\varphi}^T(k) \hat{\underline{\theta}}}$$

ARX approx.
MODEL.